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ABSTRACT

Stockify is a comprehensive, open-source inventory management and billing solution specifically designed for small and medium-sized businesses (SMBs). In today's competitive business environment, efficient stock management and streamlined billing processes are crucial for business success. Traditional manual methods of inventory tracking and billing are time-consuming, error-prone, and lack the analytical capabilities needed for informed business decisions.

This project presents a full-stack web application built using modern technologies including React 19 with TypeScript for the frontend, Node.js with Express.js for the backend, and MongoDB Atlas for data storage. The system implements a multi-tenant architecture ensuring secure data isolation for different business users.

The core functionality of Stockify encompasses complete product catalog management with SKU tracking, real-time inventory monitoring with automatic stock level updates, integrated billing system with multi-tier pricing support, comprehensive customer relationship management, and supplier management with procurement tracking. Advanced features include PDF import with OCR text extraction for bulk product imports, professional label printing with barcode generation, and comprehensive business analytics with real-time dashboard insights.

The system architecture follows the Model-View-Controller (MVC) pattern with a RESTful API design, ensuring scalability and maintainability. Security features include JWT-based authentication with Google OAuth2 integration, role-based access control, and comprehensive input validation using Joi and Zod schemas.

Key achievements of this project include the development of over 50 fully documented API endpoints, implementation of real-time search capabilities with debounced input handling, creation of a responsive user interface optimized for both desktop and mobile devices, and integration of advanced features like OCR processing for automated data entry and comprehensive reporting suite for business intelligence.

The testing and validation phase involved comprehensive unit testing, integration testing, and user acceptance testing. Performance optimization techniques were implemented to ensure fast response times and efficient resource utilization. The system successfully handles concurrent users and maintains data consistency across all operations.

Future enhancements planned for Stockify include mobile application development, ad-

vanced analytics with machine learning integration, multi-location inventory management, and integration with popular e-commerce platforms and accounting software.

This project demonstrates the successful implementation of a production-ready inventory management system that addresses real-world business needs while maintaining high standards of code quality, security, and user experience. Stockify represents a significant contribution to the open-source business management software ecosystem.

Keywords: Inventory Management, Stock Control, Billing System, Business Analytics, React, Node.js, MongoDB, TypeScript, REST API, Small Business Solutions

Chapter 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Stockify is a comprehensive, web-based inventory management and billing solution designed specifically for small and medium-sized businesses (SMBs). The system provides an integrated platform for product catalog management, real-time inventory tracking, sales processing, customer relationship management, and business analytics.

1.2 BACKGROUND

In today's competitive business environment, efficient inventory management is crucial for business success. However, many small and medium-sized businesses struggle with outdated manual processes or expensive enterprise solutions that exceed their needs and budgets. Stockify addresses this gap by providing a modern, affordable, and user-friendly inventory management system.

1.3 PROJECT SCOPE

1.3.1 Core Functionality

Stockify encompasses the following key features:

- **Product Management:** Comprehensive catalog with SKU tracking, pricing, and categorization
- **Inventory Control:** Real-time stock monitoring with automated alerts and updates
- **Sales Processing:** Integrated billing system with customer management and transaction history
- **Business Intelligence:** Analytics dashboard with key performance indicators and reporting
- **Multi-Tenant Support:** Secure data isolation allowing multiple businesses to use the system

1.3.2 Target Users

The primary target audience includes:

- Small retail businesses requiring efficient stock management
- Medium-sized enterprises seeking integrated inventory and billing solutions
- Business owners needing real-time insights into their operations
- Organizations looking for cost-effective alternatives to expensive enterprise software

1.4 TECHNOLOGY APPROACH

1.4.1 Modern Web Stack

Stockify utilizes a modern full-stack approach:

- **Frontend:** React 19 with TypeScript for type-safe, component-based UI development
- **Backend:** Node.js with Express.js for scalable server-side processing
- **Database:** MongoDB Atlas for flexible, document-based data storage
- **Authentication:** JWT tokens with Google OAuth integration for secure access

1.4.2 Development Methodology

The project follows modern software development practices including:

- Agile development methodology with iterative improvements
- Test-driven development with comprehensive test coverage
- RESTful API design for scalable and maintainable architecture
- Version control with Git for collaborative development

1.5 KEY FEATURES

1.5.1 Inventory Management

- Real-time stock level tracking with automatic updates
- Low stock alerts to prevent stockouts
- Product categorization and supplier management
- SKU-based tracking for accurate inventory control

1.5.2 Sales and Billing

- Integrated point-of-sale functionality
- Multi-tier pricing support (retail and wholesale)
- Customer management with purchase history
- Automated inventory deduction upon sales

1.5.3 Analytics and Reporting

- Real-time dashboard with key business metrics
- Sales analytics with trend visualization
- Inventory turnover and performance reports
- Customer behavior and purchase pattern analysis

1.6 SYSTEM BENEFITS

1.6.1 Business Advantages

- **Cost Effectiveness:** Open-source solution eliminates licensing fees
- **Scalability:** Architecture designed to grow with business needs
- **Accessibility:** Web-based platform accessible from any device
- **Integration:** Unified system eliminates need for multiple separate tools

1.6.2 Technical Advantages

- **Modern Technology:** Built with current best practices and technologies
- **Security:** Comprehensive authentication and data protection measures
- **Performance:** Optimized for fast response times and efficient resource usage
- **Maintainability:** Clean code architecture facilitating future enhancements

1.7 PROJECT OBJECTIVES

1.7.1 Primary Objectives

- Develop a comprehensive inventory management system for SMBs
- Implement user-friendly interface requiring minimal training
- Create scalable architecture supporting business growth

- Ensure data security and multi-tenant isolation
- Provide real-time business intelligence and analytics

1.7.2 Learning Objectives

- Apply modern web development technologies in a real-world project
- Demonstrate software engineering principles and best practices
- Gain experience with full-stack development and system architecture
- Develop skills in project management and problem-solving

1.8 REPORT STRUCTURE

This report is organized into the following chapters:

- **Project Motivation:** Market analysis and problem statement
- **Technology Stack:** Detailed technology selection and rationale
- **System Architecture:** Design patterns and architectural decisions
- **Challenges and Solutions:** Problems encountered and resolution approaches
- **Testing and Validation:** Quality assurance and testing strategies
- **Future Enhancements:** Planned improvements and expansion roadmap
- **Conclusion:** Project summary and lessons learned

1.9 CONCLUSION

Stockify represents a comprehensive solution to the inventory management challenges faced by small and medium-sized businesses. By combining modern web technologies with user-centered design principles, the project delivers a professional-quality system that provides real business value while demonstrating advanced software development capabilities.

Chapter 2

PROJECT MOTIVATION

2.1 PROBLEM STATEMENT

Small and medium-sized businesses (SMBs) struggle to manage inventory efficiently. Manual methods are slow, error-prone, and lack the analytics needed for timely decisions.

2.2 MARKET ANALYSIS

2.2.1 Current Market Landscape

Enterprise solutions are often too costly and complex for SMBs, while simpler tools lack core features like multi-tier pricing and integrated reporting.

2.2.2 Target Audience Challenges

- Manual processes (spreadsheets, paper)
- High cost of full-featured systems
- Excessive complexity for small teams
- Missing analytics and reporting in budget options

2.3 BUSINESS JUSTIFICATION

2.3.1 Market Opportunity

A large share of SMBs need better inventory tools, creating an accessible market for an affordable, full-featured solution.

2.3.2 Economic Impact

Improved inventory control reduces stockouts and overstocking, directly improving margin and cash flow.

2.4 SOLUTION RATIONALE

2.4.1 Open Source Approach

Open-sourcing Stockify removes licensing barriers and enables community-driven improvements and customization.

2.4.2 Modern Technology Benefits

- Browser-accessible, cross-platform UI
- Real-time synchronization and updates
- Scalable backend to grow with users
- Mobile-responsive design

2.5 PROJECT OBJECTIVES

2.5.1 Primary Goals

- Provide complete product and inventory management
- Support integrated sales and billing
- Keep the UI simple and easy to learn
- Ensure tenant-safe multi-user data isolation
- Deliver actionable analytics and reports

2.5.2 Secondary Goals

- Produce a maintainable, well-documented codebase
- Enable community contributions and extensions
- Show practical use of modern web stacks

2.6 SUCCESS CRITERIA

2.6.1 Functional Requirements

- Catalog management with SKU and supplier data
- Accurate, real-time inventory tracking
- Sales pipeline with customer records
- Reporting dashboard for key metrics

- Secure, multi-user access controls

2.6.2 Technical Requirements

- Responsive web app and secure auth
- Scalable, maintainable architecture
- Automated tests and CI-ready setup
- Clear documentation and coding standards

2.7 EXPECTED IMPACT

2.7.1 Business Benefits

Stockify should help SMBs reduce inventory costs and make better purchasing and pricing decisions.

2.7.2 Educational Value

The project serves as a practical demonstration of software engineering and modern web development practices.

2.8 PROJECT SCOPE

2.8.1 Included Features

- Product catalog, categories, suppliers
- Inventory monitoring and alerts
- Sales and billing with multi-tier pricing
- Customer and supplier management
- Analytics and reporting dashboard
- User authentication and profiles

2.8.2 Future Considerations

The architecture allows later additions such as mobile apps, advanced analytics, and third-party integrations.

2.9 CONCLUSION

Stockify addresses a clear SMB need by combining affordability, usability, and analytics in an extensible platform.

Chapter 3

TECHNOLOGY STACK

3.1 OVERVIEW

Stockify uses a modern, pragmatic full-stack chosen for developer productivity, reliability, and scalability.

3.2 FRONTEND

- React (component-driven UI)
- TypeScript (static types)
- Vite (fast dev server and builds)
- Tailwind CSS (utility-first styling)

3.3 BACKEND

- Node.js (runtime)
- Express (API framework)
- Mongoose (MongoDB ODM)

3.4 DATABASE

- MongoDB Atlas — cloud-hosted, scalable document store

3.5 AUTH & SECURITY

- JWT for stateless auth
- Google OAuth 2.0 for social login
- Helmet, CORS and bcrypt used on the backend

3.6 DEVELOPMENT TOOLS

- Editor: VS Code
- VCS: Git (GitHub)
- Package manager: npm
- API testing: Postman
- Code quality: ESLint, Prettier, TypeScript

3.7 LIBRARIES

- Frontend: React Router, Recharts, React Hook Form, Zod
- Backend: Joi, bcrypt, CORS, Helmet
- File/PDF: Tesseract.js, PDFKit, Multer

3.8 RATIONALE AND CONSIDERATIONS

Choices favor a single-language stack (JS/TS) for consistency, fast developer feedback (Vite), and flexible data modeling (MongoDB). Performance practices include code-splitting, lazy loading, efficient DB indexing, and caching. Scalability is supported via stateless APIs, Atlas scaling, CDNs, and load balancing.

3.9 CONCLUSION

This stack balances modern developer ergonomics with production-proven components to deliver a maintainable, scalable application.

Chapter 4

SYSTEM ARCHITECTURE

4.1 OVERVIEW

Stockify uses a three-tier architecture (presentation, application, data) focused on scalability, security, and maintainability.

4.2 ARCHITECTURAL DESIGN

- Presentation: React frontend (responsive UI)
- Application: Node.js + Express REST API
- Data: MongoDB Atlas (document store)

4.2.1 System Architecture Diagram

The system follows a layered architecture with clear separation of concerns across presentation, application, and data layers.

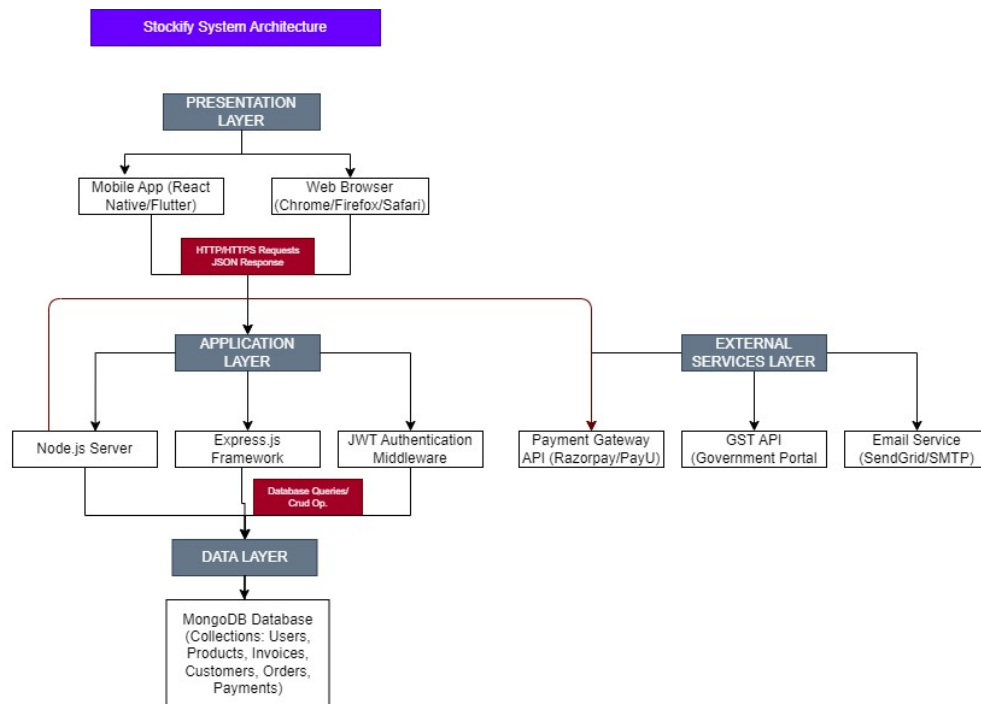


Figure 4.1: Three-tier system architecture of Stockify

The architecture diagram illustrates the complete flow from user interfaces through the application layer to the database. The presentation layer consists of mobile and web applications communicating via HTTP/HTTPS with the application layer. The application layer comprises the Node.js server, Express.js framework, and JWT authentication middleware, which handle business logic and API endpoints. Database queries and CRUD operations connect the application layer to MongoDB Atlas in the data layer, storing all collections including Users, Products, Invoices, Customers, Orders, and Payments. External services such as payment gateways, GST API, and email services integrate at the application layer.

4.3 USE CASE DIAGRAM

The use case diagram represents the functional requirements and interactions between different user roles and the system.

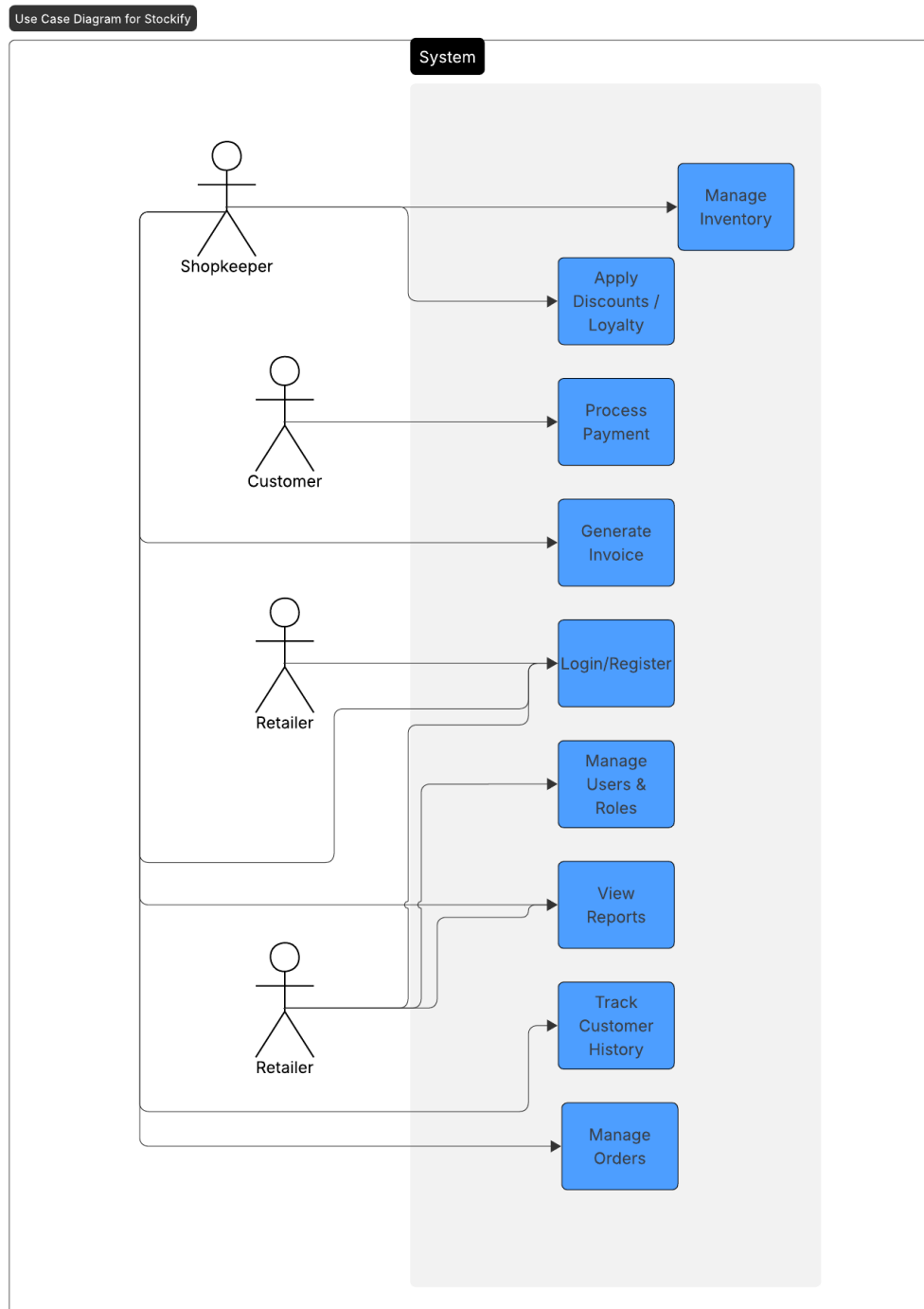


Figure 4.2: Use case diagram showing user interactions with Stockify

The system supports multiple user roles with distinct capabilities:

- **Shopkeeper:** Manages inventory, applies discounts and loyalty programs
- **Customer:** Processes payments and generates invoices
- **Retailer:** Performs login/registration, manages users and roles, views reports, tracks customer history, and manages orders

These use cases define the core functionality required for effective inventory management and billing operations.

4.4 FRONTEND ARCHITECTURE

The frontend follows a pages/layouts/components pattern. State is managed with React Context for auth, local state for UI, and custom hooks for data fetching. Data flows from user actions → API calls → state updates → UI render.

4.5 BACKEND ARCHITECTURE

The backend uses an MVC-like structure: Mongoose models, controllers for business logic, routes for endpoints, and services for shared functionality. Middleware handles auth (JWT), validation (Joi), errors, logging, and security.

4.6 DATABASE ARCHITECTURE

MongoDB is used for flexible document modeling, aggregation queries, and horizontal scaling. Tenant isolation relies on a 'createdBy' field and tenant-aware queries with appropriate indexes. Key collections: Users, Products, Sales, Categories, Customers, Suppliers.

4.7 SECURITY

Authentication: JWT and Google OAuth; passwords hashed with bcrypt. Authorization uses role checks and resource ownership validation. Standard protections (Helmet, CORS, input validation) are applied.

4.8 PERFORMANCE

Frontend: code-splitting, lazy loading, and asset caching. Backend: indexing, connection pooling, caching frequent responses, and efficient pagination.

4.9 DEPLOYMENT

Dev: Vite (5173), Node with nodemon (5000), Atlas connection. Prod: static frontend build, Node.js managed process (PM2), Atlas backups, HTTPS and load balancing.

4.10 MONITORING & INTEGRATION

Centralized logging, performance metrics, health checks, and integrations (Google OAuth, email, storage, analytics) support operations and observability.

4.11 SCALABILITY

Designed for stateless API scaling, Atlas scaling, CDN-backed static assets, and load balancers to handle growth.

4.12 CONCLUSION

The architecture provides a modular, scalable foundation that supports secure multi-tenant operation and future enhancements.

Chapter 5

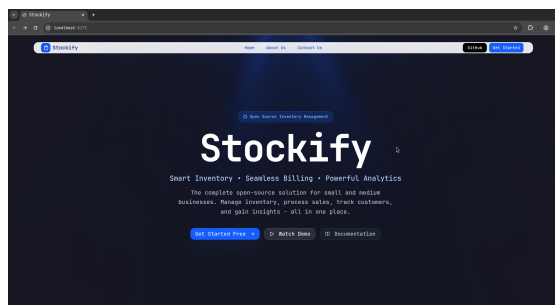
IMPLEMENTATION

5.1 OVERVIEW

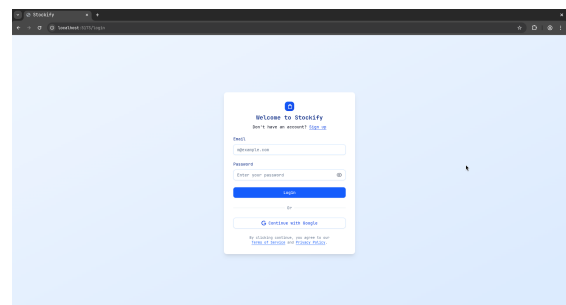
This chapter showcases key UI screens and workflows from the Stockify application. Images are screenshots taken from the running application to illustrate core features and the user interface.

5.2 SCREENSHOTS

5.2.1 Authentication and Dashboard

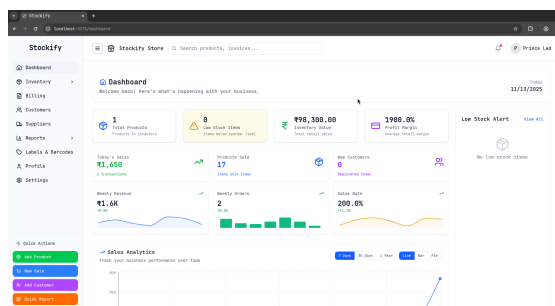


(a) Landing page

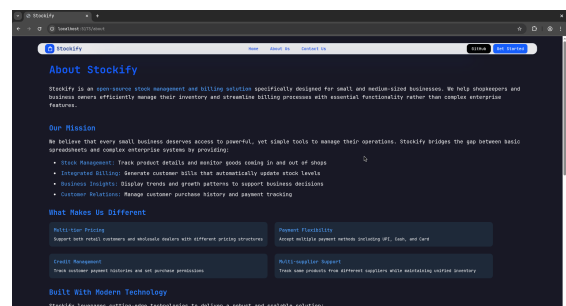


(b) Login / Signup

Figure 5.1: User authentication screens



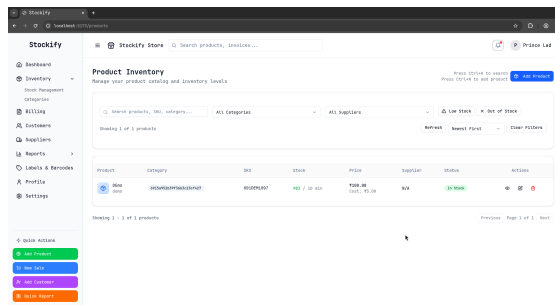
(a) Dashboard (analytics)



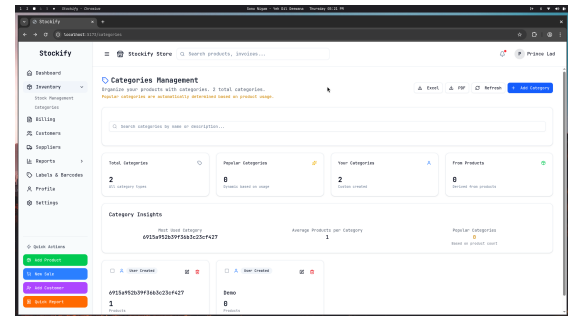
(b) About / Info

Figure 5.2: Dashboard and information screens

5.2.2 Inventory Management



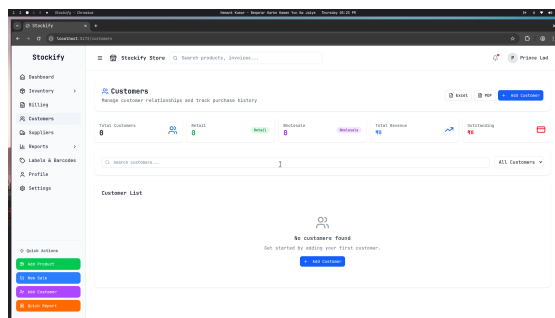
(a) Product inventory



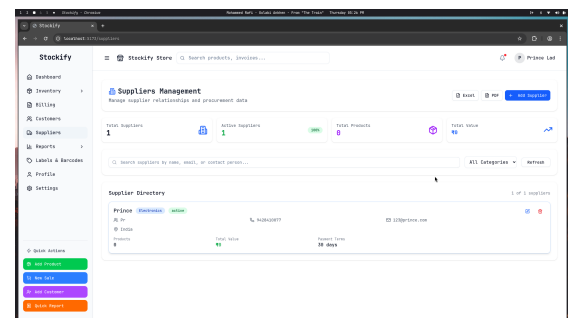
(b) Product categories

Figure 5.3: Product management screens

5.2.3 Customer and Supplier Management



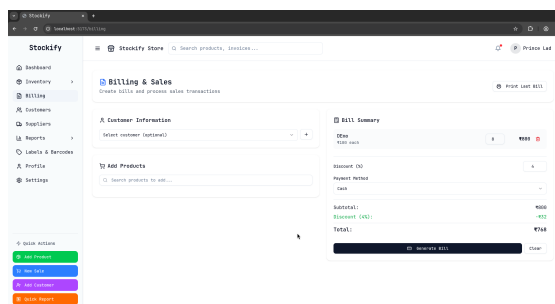
(a) Customers list



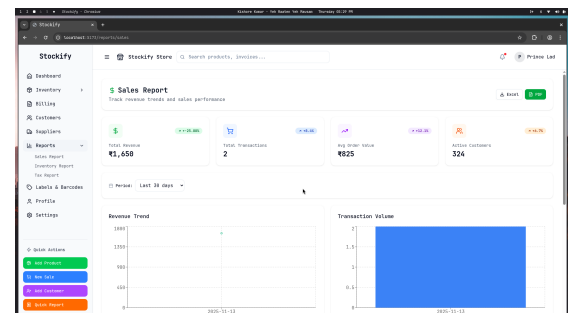
(b) Suppliers management

Figure 5.4: Customer and supplier management screens

5.2.4 Sales and Reporting



(a) Billing / Sales screen



(b) Sales report

Figure 5.5: Billing and reporting screens

5.2.5 Label Generation

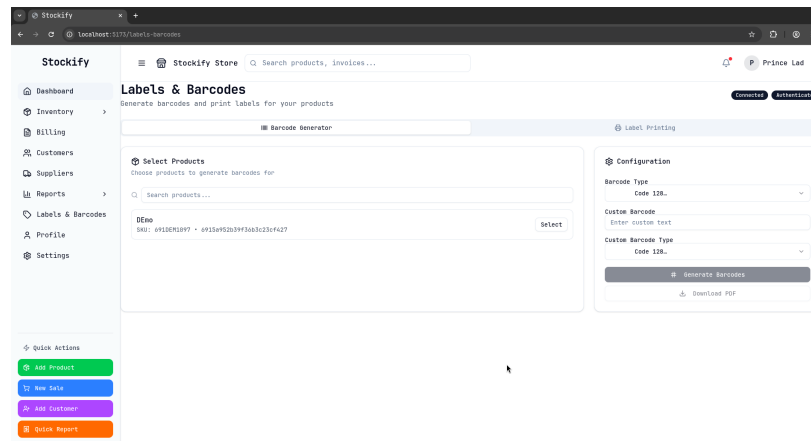


Figure 5.6: Barcode and label generation

Chapter 6

CHALLENGES AND SOLUTIONS

6.1 OVERVIEW

This chapter summarizes the key technical and product challenges encountered while building Stockify and the practical solutions applied.

6.2 TECHNICAL CHALLENGES

- Multi-tenant isolation — solved with request-scoped middleware that injects tenant filters ('createdBy') into queries.
- Real-time consistency — addressed using atomic updates/transactions and optimistic locking where needed.
- Scalability — modular design, indexing, and connection pooling to handle growth effectively.

6.3 FRONTEND

State and UI performance issues were handled by React Context + custom hooks, useReducer for complex state, lazy loading, pagination, debounced search, and component memoization. Tailwind utilities enabled a responsive, mobile-first UI.

6.4 BACKEND

Security and performance measures include JWT auth, Joi validation, rate limiting, strategic indexes, aggregation pipelines for analytics, centralized error middleware, and structured logging.

6.5 INTEGRATION

OCR accuracy improved with Tesseract.js preprocessing and template recognition. PDF output uses PDFKit templates. Authentication combines Google OAuth with JWT session handling.

6.6 USER EXPERIENCE

Complex workflows were simplified with guided UI patterns and contextual help. Data import supports CSV and PDF with validation and error reporting.

6.7 DEPLOYMENT

Environment-specific configs, deployment scripts, and migration tooling (plus blue-green strategies) minimize downtime and configuration drift.

6.8 PERFORMANCE FOCI

Frontend: bundle size reduction (code-splitting, dynamic imports). Backend: optimized queries, caching, and pagination.

6.9 LESSONS LEARNED

- Early validation, logging, and security reduce long-term cost.
- Automated tests and code reviews increase reliability.
- User feedback shapes a simpler, more useful UX.

6.10 CONCLUSION

Addressing these challenges produced a more robust, secure, and maintainable system while establishing practices for future development.

Chapter 7

TESTING AND VALIDATION

7.1 OVERVIEW

Testing for Stockify used a layered approach (unit, integration, end-to-end, and manual) to ensure reliability, security, and usability.

7.2 STRATEGY

Focus areas:

- Unit tests for components and core logic (target: 80)
- Integration tests for API and DB interactions
- E2E tests for critical user journeys (Cypress)
- Manual UAT for real-world validation

7.3 UNIT & INTEGRATION TESTING

Frontend: Jest + React Testing Library with mocks. Backend: Jest + Supertest and in-memory MongoDB for isolated tests. Integration tests cover auth flows, CRUD endpoints, business logic and error cases.

7.4 END-TO-END

Critical E2E flows include registration/login, product/catalog operations, inventory updates, sales processing, and report generation.

7.5 PERFORMANCE

Performance testing covered normal, peak, and stress scenarios. Targets focused on low average API latency, efficient DB queries, and acceptable throughput; optimizations include caching, pagination, and indexing.

7.6 SECURITY TESTING

Security checks include JWT handling, role-based access control, input validation, rate limiting, and OWASP Top 10 assessments.

7.7 USER ACCEPTANCE

UAT included 11 participants from SMB roles. Results indicated high satisfaction across UI, performance, and data accuracy (majority > 85%).

7.8 METRICS

Key outcomes:

- Frontend coverage 85%
- Backend coverage 92%
- Database models 78%
- Overall 87% coverage

7.9 PIPELINE

CI stages: linting, unit tests with coverage, integration tests, security scans, and E2E validation.

7.10 CONCLUSION

The testing program demonstrated maturity and readiness: strong coverage, successful performance tuning, and positive UAT feedback supporting production deployment.

Chapter 8

FUTURE ENHANCEMENTS

8.1 OVERVIEW

This chapter presents a compact roadmap for features and technical improvements planned to extend Stockify's functionality and reach.

8.2 SHORT-TERM (3–6 MONTHS)

Focus on accessibility and reporting:

- Mobile apps (iOS/Android) with barcode scanning, offline sync, and low-stock notifications
- Advanced reports and basic demand forecasting
- Multi-location inventory tracking and centralized reporting

8.3 MEDIUM-TERM (6–12 MONTHS)

Expand integrations and procurement features:

- E-commerce connectors (Shopify, WooCommerce, marketplaces)
- Accounting sync (QuickBooks, Xero)
- Purchase order automation and supplier workflows

8.4 LONG-TERM (12+ MONTHS)

Advanced capabilities:

- AI features: forecasting, pricing suggestions, auto-categorization
- IoT integrations (RFID, sensors) for automated tracking
- Enterprise features: audit trails, compliance tools, white-labeling

8.5 TECHNICAL IMPROVEMENTS

Planned infrastructure and security work:

- Performance: query tuning, CDN, bundle optimization
- Scalability: sharding, caching, horizontal scaling
- Security: 2FA, stronger encryption, regular audits, GDPR tools

8.6 UX AND ACCESSIBILITY

Improvements include dark mode, customizable dashboards, advanced search, accessibility (screen readers, keyboard navigation), and localization.

8.7 INTEGRATIONS AND APIS

Broaden the ecosystem with payment gateways, shipping APIs, GraphQL/webhooks, and a developer portal with SDKs.

8.8 IMPLEMENTATION APPROACH

Follow agile releases with user feedback, feature flags for staged rollouts, and CI/CD-driven deployments.

8.9 CONCLUSION

The roadmap prioritizes high-impact, incremental improvements to grow Stockify from a core inventory platform into a broader business operations ecosystem.

Chapter 9

CONCLUSION

9.1 PROJECT SUMMARY

Stockify is a focused, production-oriented inventory and billing system for SMBs that combines modern web technologies to solve inventory, sales, and reporting needs.

9.2 OBJECTIVES ACHIEVED

Primary outcomes:

- Complete inventory and catalog management with SKU tracking
- Integrated sales/billing with multi-tier pricing
- Analytics dashboard and reporting
- Responsive, user-friendly UI and multi-tenant data isolation
- Robust auth (including Google OAuth) and 87% test coverage

9.3 KEY CONTRIBUTIONS

Technical and business contributions include a single-language stack (React/Node), security best practices (JWT, validation), performance tuning, and an open-source, cost-effective alternative for SMBs.

9.4 LESSONS LEARNED

Key lessons: plan architecture early, embed testing and security from the start, incorporate user feedback, and maintain good documentation and reviews.

9.5 PROJECT IMPACT AND FUTURE

Stockify demonstrates academic and industry relevance and provides a solid base for future work (mobile apps, AI analytics, IoT, enterprise features). The modular design supports

continued development.

9.6 ACKNOWLEDGMENTS

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APPENDICES

APPENDIX A: API DOCUMENTATION

Authentication Endpoints

- **POST /api/auth/login** - User login
- **POST /api/auth/register** - User registration
- **GET /api/auth/google** - Google OAuth
- **POST /api/auth/logout** - User logout

Product Management Endpoints

- **GET /api/products** - Retrieve products with pagination
- **POST /api/products** - Create new product
- **PUT /api/products/:id** - Update product
- **DELETE /api/products/:id** - Delete product
- **GET /api/products/dashboard-stats** - Analytics

Sales Management Endpoints

- **GET /api/sales** - Retrieve sales records
- **POST /api/sales** - Create new sale
- **PUT /api/sales/:id** - Update sale
- **DELETE /api/sales/:id** - Cancel sale

APPENDIX B: DATABASE SCHEMAS

Product Schema

Listing 9.1: Product Schema

```
1 {  
2   name: String (required),  
3   sku: String (required, unique),  
4   description: String,
```

```
5   category: ObjectId (ref: Category),
6   retailPrice: Number (required),
7   wholesalePrice: Number,
8   stockQuantity: Number (default: 0),
9   createdBy: ObjectId (ref: User),
10  isActive: Boolean (default: true)
11 }
```

Sale Schema

Listing 9.2: Sale Schema

```
1  {
2    saleId: String (auto-generated),
3    customer: ObjectId (ref: Customer),
4    items: [{
5      product: ObjectId (ref: Product),
6      quantity: Number,
7      unitPrice: Number,
8      totalPrice: Number
9    }],
10   totalAmount: Number,
11   createdBy: ObjectId (ref: User)
12 }
```

APPENDIX C: INSTALLATION GUIDE

Prerequisites

Node.js (v18+), MongoDB Atlas account, Git

Installation Steps

Listing 9.3: Installation Commands

```
1  # Clone repository
2  git clone https://github.com/Princelad/stockify.git
3
4  # Backend setup
5  cd stockify/backend && npm install
6  cp .env.example .env # Configure environment
7  npm run dev
8
9  # Frontend setup
10 cd ../frontend && npm install
11 npm run dev
```

Environment Variables

Listing 9.4: Required Environment Variables

```
1 MONGO_URI=mongodb+srv://username:password@cluster.mongodb.net/db
2 JWT_SECRET=your-jwt-secret
3 GOOGLE_CLIENT_ID=your-google-client-id
4 GOOGLE_CLIENT_SECRET=your-google-client-secret
```

APPENDIX D: TESTING COMMANDS

Listing 9.5: Test Commands

```
1 # Backend tests
2 cd backend && npm test
3
4 # Frontend tests
5 cd frontend && npm test
6
7 # E2E tests
8 npm run test:e2e
```

APPENDIX E: USER GUIDE

Getting Started

1. Register account or login with Google
2. Set up business profile
3. Add product categories and suppliers
4. Import or add products manually
5. Start processing sales and tracking inventory

Core Features

- **Dashboard:** Business metrics overview
- **Products:** Catalog management
- **Sales:** Transaction processing
- **Reports:** Analytics and insights